5. FREKVENCIJSKI ODZIV POJAČALA

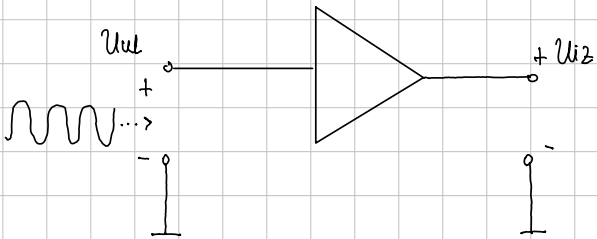
Frekvencijski odziv

- pri utjecaju reaktancija → prijenos fje pojačala je kompleksni br.

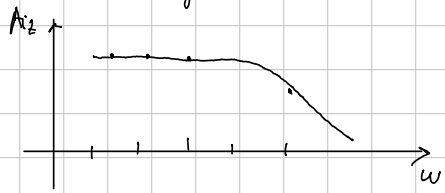
Frekvencijski odziv → odziv pojačala na sinus signale različitih frekv. uz linearan rad pojačala

$U_{ul} = U_{ulM} \cdot \sin \omega t$ $\qquad$ $U_{iz} = U_{izM} \sin(\omega t + \alpha)$

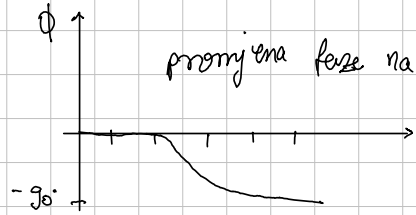
za svaku frekvenciju $f = \omega / 2\pi$ određuju se AMPLITUDA i FAZA



mijenjamo frekvenciju i mjerimo Av



$$|A_v(f)| = \frac{U_{izM}}{U_{ulM}} \qquad \measuredangle A_v(f) = \phi$$

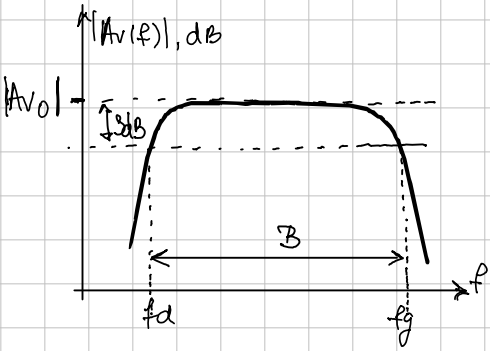


promjena faze na izlazu

naponsko pojačanje = complex br → $A_v(f) = |A_v(f)| \cdot \exp(j\phi)$

graf prikaz frekv odziva: frekvencijska karakteristika

Amplitudna frekv. karakt. izmjeničnog pojačala



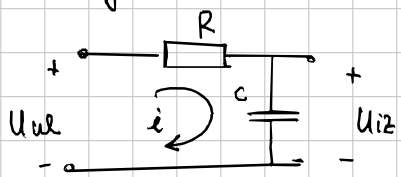
Granične frekv. - pad pojačanja na $A_{v0}/\sqrt{2}$ ili za 3dB

$$A_{v_{dB}} = 20 \log |A_v| \quad [dB], \quad A_v = \frac{U_{iz}}{U_{ul}}, \quad T = \frac{U_{iz}}{U_{ul}}$$

· pojačanje se smanjuje u području niskih frekv. i u području visokih frekv.

Polovi i nule prijenosne funkcije ⟹ TO ZNAŠ IZ ELEKTRE

Primjer:



$$T = \frac{U_{iz}}{U_{ul}} = \frac{i \cdot Z_C}{i \cdot (R + Z_C)} = \frac{Z_C}{R + Z_C}$$

$$Z_C = \frac{1}{j\omega C} \implies T = \frac{\frac{1}{j\omega C}}{R + \frac{1}{j\omega C}} \quad \Big| \cdot j\omega C$$

$$T = \frac{1}{j\omega RC + 1}$$ → moramo dobiti Re i Im

$$\implies T = Re\{\} + Im\{\}$$

$$\tau = R \cdot C \implies \omega_1 = \frac{1}{\tau} = \frac{1}{RC}$$

(karakteristična kružna frekv.)

$$\implies T = \frac{1}{j\frac{\omega}{\omega_1} + 1}$$

Kada bismo tražili polove: $s = j\omega$

$$T = \frac{1}{1 + \frac{s}{\omega_1}}$$

→ nula → nema

→ polovi: $1 + \frac{s}{\omega_1} = 0 \qquad \frac{s}{\omega_1} = -1$

ω_{p1} ⟹ $T = \frac{1}{1 + \frac{s}{\omega_{p1}}}$

→ prvi pol

Bodeov dijagram

• amplitudna i fazna karakteristika u ovisnosti o frekvenciji

AMPLITUDA : [dB]

FAZA : [°] (stupnjevi)

FREKVENCIJA : log mjerilo

za crtanje gledamo frekvencijski oblik

$$\rightarrow T = \frac{1}{1+\frac{\omega}{\omega_1}j}$$

$$\dashrightarrow T = \frac{1}{1+\frac{\omega}{\omega_i}j} \qquad |T| = ? \ [dB] \qquad \phi = ?$$

trebamo neki oblik kojem možemo odrediti Re i Im

$$|T| = \sqrt{(Re\{\cdot\})^2 + (Im\{\cdot\})^2} = \sqrt{\frac{1}{\left(1+\left(\frac{\omega}{\omega_i}\right)^2\right)^2} + \frac{\left(\frac{\omega}{\omega_i}\right)^2}{\left(1+\left(\frac{\omega}{\omega_i}\right)^2\right)^2}}$$

$$T = \frac{1\cdot\left(1-\frac{\omega}{\omega_i}j\right)}{1-\left(-\frac{\omega^2}{\omega_i^2}\right)} = \underbrace{\frac{1}{1+\frac{\omega^2}{\omega_i^2}}}_{Re} - j\underbrace{\frac{\frac{\omega}{\omega_i}}{1+\frac{\omega^2}{\omega_i^2}}}_{Im}$$

! ne uvrštavaš !

$$\Downarrow$$

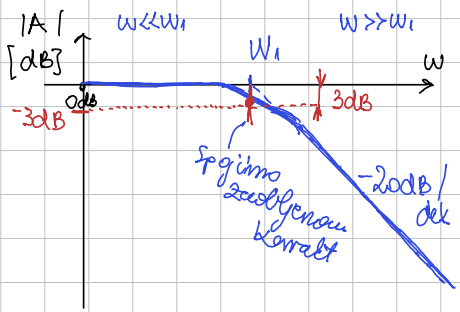
$$\sqrt{\frac{1}{1+\left(\frac{\omega}{\omega_i}\right)^2}}$$

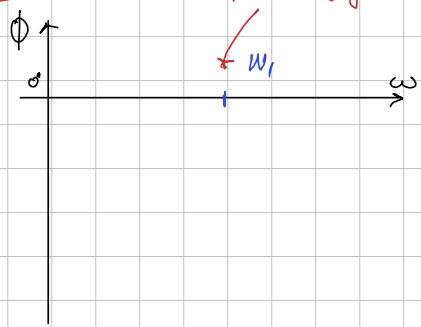
$$\Rightarrow |T| = \frac{1}{\sqrt{1+\left(\frac{\omega}{\omega_i}\right)^2}} \rightarrow |T|_{dB} = -20\log\left(\sqrt{1+\left(\frac{\omega}{\omega_i}\right)^2}\right)$$

kad bismo to htjeli crtati koristili bismo Bodeov dijagram

$\omega_i = \frac{1}{RC} = konst$

ω mijenjamo i onda nacrtamo





$|T|: \omega \ll \omega_1 \rightarrow |T| \approx 1$

$= 20\log 1 = 0$ dB

$: \omega \gg \omega_1 \Rightarrow |T| \approx \frac{1}{\frac{\omega}{\omega_1}} \rightarrow$ pada

$= 20$ dB $\rightarrow -20$ dB

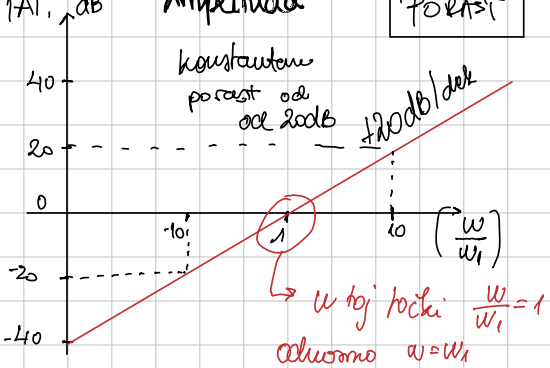
Osnovni član - NULA na frekvenciji 0

$$A(j\omega) = j\frac{\omega}{\omega_1}$$

$$\rightarrow |A| = \sqrt{Re\{\cdot\}^2 + Im\{\cdot\}^2} = \sqrt{0 + \left(\frac{\omega}{\omega_1}\right)^2} \Rightarrow |A| = \left|\frac{\omega}{\omega_1}\right|$$

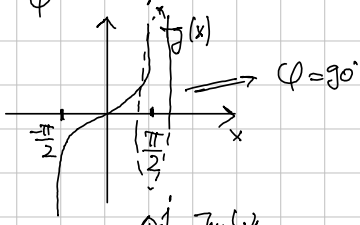
Amplituda $\boxed{\text{PORAST}}$

$$\Rightarrow |A|_{dB} = 20\log\left(\frac{\omega}{\omega_1}\right) \quad [dB]$$

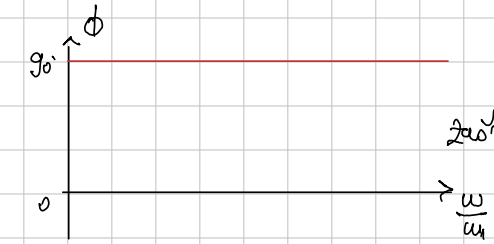


Faza: $\phi = \operatorname{arctg}\frac{Im\{\cdot\}}{Re\{\cdot\}} = \frac{\frac{\omega}{\omega_1}}{0}$

$\phi = \operatorname{arctg}\infty$

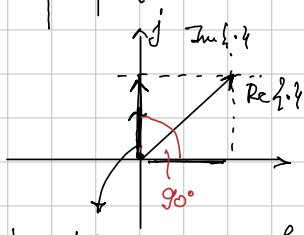


$\Rightarrow \varphi = 90°$



zašto je konst?

$\Rightarrow$



bez obzira na promjenu frekv (pa se smanjuje / povećava na gori) ne dolazi do promjene faze jer $Re\{\cdot\} = 0$ pa uvijek zahvaća isti kut

Osnovni član - POL na frekvenciji 0

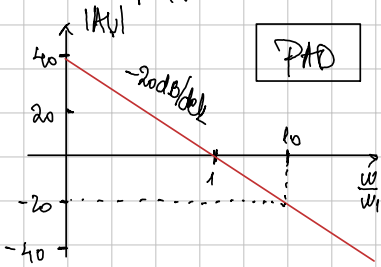
$$A(j\omega) = \frac{1}{j\frac{\omega}{\omega_1}} \quad \begin{matrix} /\cdot j\frac{\omega}{\omega_1} \\ /\cdot j\frac{\omega}{\omega_1} \end{matrix} = \frac{j\frac{\omega}{\omega_1}}{\left(\frac{\omega}{\omega_1}\right)^2} = 0 - j\frac{1}{\frac{\omega}{\omega_1}} = 0 + j\left(-\frac{1}{\frac{\omega}{\omega_1}}\right)$$

$$|A| = \left|\frac{1}{\frac{\omega}{\omega_1}}\right| \longrightarrow |A|_{dB} = 20\log(\overset{=0}{1}) - 20\log\left(\frac{\omega}{\omega_1}\right) \quad [dB] = -20\log\left(\frac{\omega}{\omega_1}\right)$$

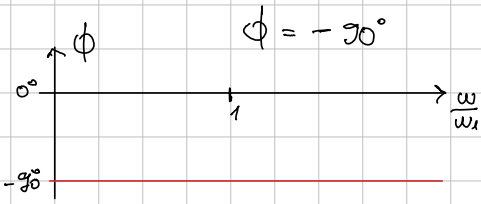
$$|A| = \sqrt{\left(\frac{\omega_1}{\omega}\right)^2} \qquad \omega\uparrow \sim |A|\downarrow$$

$$\phi = \operatorname{arctg}\frac{-\frac{\omega}{\omega_1}}{0} = -\infty$$

$$\phi = -90°$$

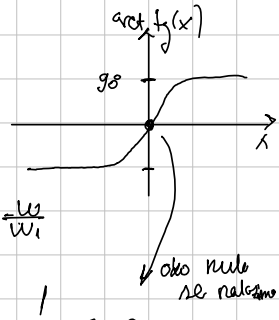


$\boxed{\text{PAD}}$



Osnovni član – NULA na frekvenciji ω_1

$$A(j\omega) = 1 + j\frac{\omega}{\omega_1}$$



amplituda

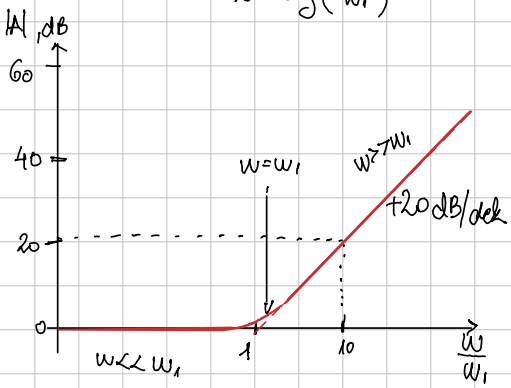
$$|A| = \sqrt{1+\left(\frac{\omega}{\omega_1}\right)^2}$$

$$|A|_{dB} = 20\log\left(\sqrt{1+\left(\frac{\omega}{\omega_1}\right)^2}\right)$$

$\omega \ll \omega_1 \longrightarrow |A| \approx 0$

$\omega = \omega_1 \longrightarrow 20\log\sqrt{2}$

$\omega \gg \omega_1 \longrightarrow 20\log\left(\frac{\omega}{\omega_1}\right)$



faza

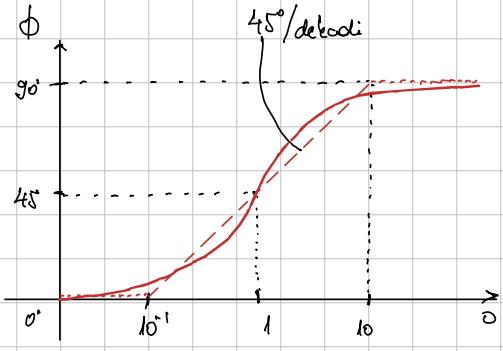
$$\phi = \operatorname{arctg}\frac{\mathrm{Im}}{\mathrm{Re}} = \operatorname{arctg}\frac{\omega}{\omega_1}$$

$\omega \ll \omega_1 \longrightarrow \operatorname{arctg}\frac{1}{\omega_1} \approx 0$

$\omega = \omega_1 \longrightarrow \phi = 45°$

$\omega \gg \omega_1 \longrightarrow \phi \approx 90°$

↳ udaljavanjem od ishodišta idemo prema 90°



Osnovni član – POL na frekvenciji ω_1

$$A(j\omega) = \frac{1}{1+j\frac{\omega}{\omega_1}} \frac{/\cdot\left(1-j\frac{\omega}{\omega_1}\right)}{/\cdot\left(1-j\frac{\omega}{\omega_1}\right)} = \frac{1-j\frac{\omega}{\omega_1}}{1+\left(\frac{\omega}{\omega_1}\right)^2} = \frac{1}{1+\frac{\omega^2}{\omega_1^2}} - j\frac{\frac{\omega}{\omega_1}}{1+\left(\frac{\omega}{\omega_1}\right)^2}$$

amplituda

$$|A| = \dots = \frac{1}{\sqrt{1+\left(\frac{\omega}{\omega_1}\right)^2}}$$

$\omega \ll \omega_1 \dashrightarrow |A| \approx 0$

$\omega = \omega_1 \dashrightarrow |A| = -20\log(\sqrt{2})$

$\omega \gg \omega_1 \dashrightarrow |A| = -20\log\left(\frac{\omega}{\omega_1}\right)$

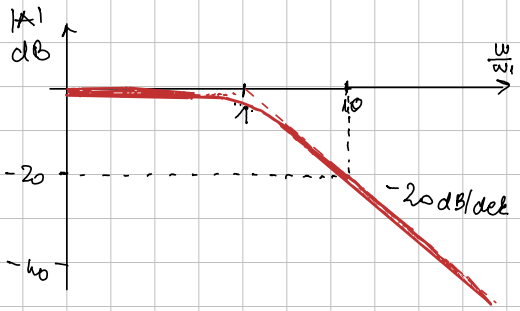
faza

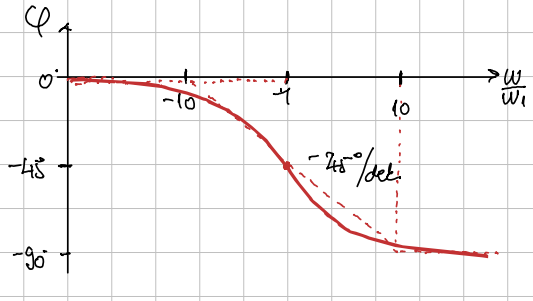
$$\varphi = \operatorname{arctg}\frac{-\frac{\omega}{\omega_1}\cancel{\frac{1}{1+\left(\frac{\omega}{\omega_1}\right)^2}}}{\cancel{\frac{1}{1+\left(\frac{\omega}{\omega_1}\right)^2}}} = \operatorname{arctg}\left(-\frac{\omega}{\omega_1}\right)$$

$\omega \ll \omega_1 \longrightarrow \varphi \approx 0$

$\omega = \omega_1 \longrightarrow \varphi \approx -45°$

$\omega \gg \omega_1 \longrightarrow \varphi \approx -90°$





Primjer 5.1.) Nacrtati Bodeov dijagram prijenosne fje

$$A(j\omega) = 10^6 \frac{j\omega}{(10^3 + j\omega)(10^5 + j\omega)} = \frac{10^6}{1} \frac{j\omega}{10^3\left(1 + \frac{j\omega}{10^3}\right) \cdot 10^5\left(1 + \frac{j\omega}{10^5}\right)}$$

$$= \frac{10^6}{10^8} \cdot \frac{j\omega}{\left(1 + \frac{j\omega}{10^3}\right)\left(1 + j\frac{\omega}{10^5}\right)} = \frac{1}{10^2} \cdot \frac{j\omega}{\left(1 + \frac{j\omega}{10^3}\right)\left(1 + j\frac{\omega}{10^5}\right)}$$

treba biti oblika $j\frac{\omega}{\omega_1}$

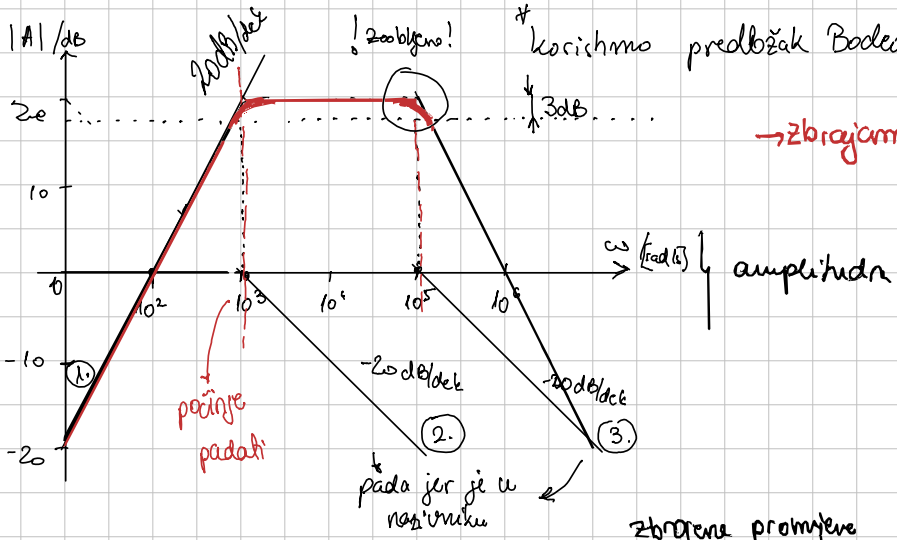
$\left(1 + j\frac{\omega}{\omega_1}\right)$

$$A(j\omega) = \frac{j\frac{\omega}{10^2}}{\left(1 + \frac{j\omega}{10^3}\right)\left(1 + j\frac{\omega}{10^5}\right)}$$

1. $j\frac{\omega}{10^2} \longrightarrow$ 1.

2. $\left(1 + \frac{j\omega}{10^3}\right)$

3. $\left(1 + j\frac{\omega}{10^5}\right)$



* korisimo predložak Bodeovog dijagrama

→ zbrajamo karakt!

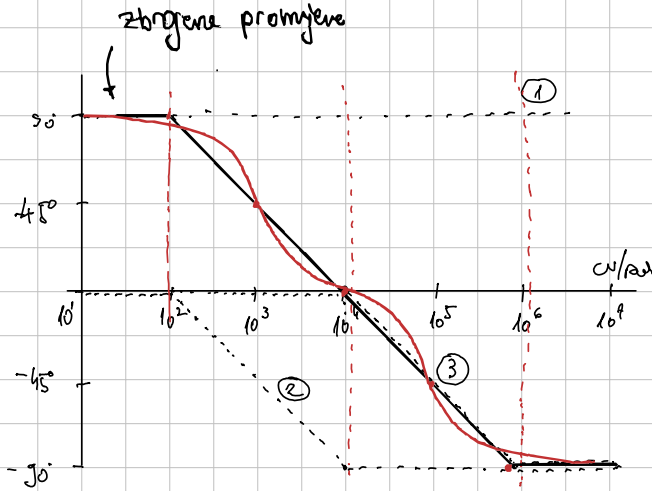
amplituda

$$A(j\omega) = \frac{\overbrace{j\frac{\omega}{10^2}}^{1.}}{\underbrace{\left(1 + \frac{j\omega}{10^3}\right)}_{2.}\underbrace{\left(1 + j\frac{\omega}{10^5}\right)}_{3.}}$$

→ 90°

—||—

promjena oko 10^3

poč: 10^2 ($10^3/10$)

kraj: 10^4 ($10^3 \cdot 10$)